

Biology Midterm — Student WD_S10

The plasma membrane is a selectively permeable, fluid phospholipid bilayer that regulates transport into and out of the cell. Small nonpolar molecules such as O₂ and CO₂ diffuse passively across the bilayer, while larger or charged solutes require facilitated diffusion or ATP-dependent active transport against their concentration gradient. Integral membrane proteins serve as receptors mediating signal transduction, and cholesterol embedded within the bilayer modulates membrane fluidity and preserves structural integrity across a range of temperatures.